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Deposit Inflows and Outflows in Failing Banks: The Role of Deposit Insurance

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1 Big picture

- **Question.** What happens to **deposit outflows and inflows** when a bank becomes distressed, and how much does **deposit insurance** stabilize funding? In particular, can a failing bank lose uninsured deposits and simultaneously replace them with new insured deposits?
- **Setting.** The paper combines a uniquely detailed **daily, account-level dataset** from a failed U.S. bank with broader evidence from **all U.S. banks facing regulatory action** and with **RateWatch** deposit-rate data.
- **Core challenge.** Most datasets and most of the bank-run literature focus on **net deposit changes** or on **outflows only**. But a bank's balance sheet can change dramatically through **gross inflows** even while uninsured depositors are running.
- **Main contribution.** The paper shows that:
 1. uninsured depositors do run after bad regulatory news;
 2. regular deposit insurance and temporary crisis-era guarantees reduce outflows;
 3. gross **insured inflows** are quantitatively first-order for a failing bank;
 4. these inflows are attracted by **above-market rates** and tend to sit just below the insurance limit;
 5. similar compositional shifts appear at a broad sample of U.S. banks facing regulatory action.
- **Why it matters.** If a troubled bank can easily replace fleeing uninsured deposits with new insured deposits, then classic depositor discipline is weakened. This has implications for bank stability, deposit-insurance design, brokered-deposit rules, and deposit-rate caps.
- **Main empirical question.** In a distressed bank, who withdraws, who stays, who newly enters, and how do deposit insurance and deposit rates shape these flows?
- **Main empirical results.**
 - Uninsured depositors are much more likely to withdraw after adverse bank-specific news.
 - Checking accounts, direct-deposit accounts, and longer relationships are more stable.
 - Temporary guarantees such as **TAG** are effective in reducing outflows.
 - In the bank's final year, it attracted nearly **\$400 million** of new insured term deposits, about **one-third of its deposit base**, roughly offsetting fleeing deposits.
 - About **half** of these new deposits arrived in the final **90 days** before failure.
 - These inflows were mostly institutional CDs, largely from **banks and credit unions**, structured just below the insurance limit and gathered through **listing services**.
 - Across U.S. banks under regulatory action, brokered deposits fall, but listing-service deposits and insured term deposits below \$250,000 rise.
 - Large banks facing enforcement actions also raise deposit rates, suggesting that the inflow mechanism is not unique to the failed bank in the microdata.
- **Economic takeaway.** Deposit insurance does not just reduce runs. It can also sustain failing banks through **new insured inflows**. To understand bank fragility, one must study **gross inflows and gross outflows separately**.

2 Data and measurement (key objects)

2.1 Main microdata from the failed bank

- **Source.** The core dataset was assembled by the FDIC after the bank failed and contains **daily account-level balances and transactions** from early 2006 until failure.
- **Granularity.** The data allow the authors to separately observe:
 - transaction versus term deposits,
 - insured versus uninsured balances,
 - extant versus new depositors,
 - depositor type (individual, trust, institutional, brokered/placed, listed/faxed),
 - account rates and fees,
 - transaction intensity,
 - direct-deposit activity,
 - account age and, for CDs, time to maturity.
- **Validation.** The constructed balances are checked against **Call Reports** and the FDIC **Summary of Deposits**, which gives confidence that the account-level data are complete for the studied bank.

2.2 The bank and the event periods

- **Bank background.** The bank had roughly **\$2 billion** in assets, focused primarily on residential real-estate lending, and also offered wealth-management services.
- **Economic timeline.** The analysis is organized around four periods:
 1. **Placebo:** a period in 2006, when neither the bank nor the financial system appeared especially troubled;
 2. **Precrisis:** roughly the year before September 2008, when stress was rising and uninsured balances were already beginning to run;
 3. **Postcrisis:** from December 2008 to May 2009, after the major emergency policy interventions, capturing system-wide stress but not bank-specific bad news;
 4. **Formal Enforcement Action:** from the public cease-and-desist order to bank failure, capturing severe **bank-specific distress**.
- **Critical undercapitalization.** In the final months, public filings showed the bank to be significantly and then critically undercapitalized, making failure within 90 days highly likely under Prompt Corrective Action rules.

2.3 Key depositor and account objects

- **Liquidation.** An account is treated as having liquidated if its balance falls by at least **75%** relative to the start of the period and remains at or below **25%** of the starting balance for at least **61 days**.

- **Insurance status.** The authors separately track whether accounts are uninsured and whether they are **TAG/DFA eligible**. TAG and DFA expanded effective coverage during the crisis.
- **Relationship-depth measures.**
 - **Log(Age):** age of the relationship/account,
 - **Prior Transactions:** fraction of days in the prior year with at least one depositor-driven transaction,
 - **Direct Deposit:** regular payroll-like inflows.
- **Wholesale funding categories.**
 - **Institutional-Listed/Faxed:** institutional deposits sourced through internet listing services or faxed requests;
 - **Institutional-Other:** institutional deposits not clearly tied to listing services;
 - **Brokered/Placed:** deposits placed by a broker or fiduciary.
- **Pricing variable.** The **Interest Rate Spread** is the account’s offered rate relative to the market average. For term deposits, the spread is constructed using newly opened same-maturity CDs in a centered rolling window and market rates from **RateWatch**.

2.4 Descriptive facts from Table I

- New accounts by period are:
 - Placebo: **3,028**
 - Precrisis: **2,146**
 - Postcrisis: **927**
 - Formal: **2,600**
- The share of uninsured new accounts falls from **6.3%** in the Placebo period to **2.7%** in the Formal period.
- Average opening balances of new depositors rise sharply over time, from about **\$43,520** in the Placebo period to about **\$186,900** in the Formal period.
- The share of **term deposits** among new accounts rises to **82.4%** in the Formal period.
- Institutional listed/faxed deposits among new accounts jump from essentially zero to **70.7%** in the Formal period.
- Among extant depositors, uninsured balances fall sharply after the crisis and by the failure date only about **1.4%** of accounts still have any uninsured funds.

2.5 Broader datasets for generalization

- **Call Reports.** Used to study the funding composition of all U.S. banks between 2000 and 2016.
- **Regulatory action data.** Public and confidential supervisory data are used to define a treatment indicator for banks under formal action or similar restrictions.

- **RateWatch.** Used to study deposit rates offered by banks, including large banks facing enforcement actions.

3 Models and empirical specifications (all equations + interpretation)

3.1 Outflow regressions at the account level

For each period and deposit type, the authors estimate a liquidation regression of the form

$$1(\text{Liquidation}_i) = \alpha + \beta \text{Controls}_i + \delta_b + \varepsilon_i.$$

- $1(\text{Liquidation}_i)$ is an indicator for whether account i runs off by the paper's 75%/61-day rule.
- Controls_i includes insurance status, TAG/DFA eligibility, account type, direct deposit, account age, prior transactions, trust status, maturity, institutional type, and rate spread.
- δ_b is a branch fixed effect.

Interpretation. This is the paper's main reduced-form design for asking who withdraws and who stays. It exploits differences across accounts exposed to different coverage, account types, and relationship depth, within each relevant stress period.

3.2 Alternative outflow specifications

The paper checks robustness using **Probit** and **Cox proportional hazard** models for the crisis periods.

Interpretation. The purpose is not to change the economic question but to verify that the findings are not an artifact of the linear probability model or of the exact treatment of liquidation timing.

3.3 Characteristics of new versus extant depositors

To understand inflows, the paper classifies how **new depositors** differ from **extant depositors** across time periods using regressions that interact period dummies with depositor status.

A generic representation is

$$1(\text{Characteristic}_{i,t}) = \alpha + \sum_p \beta_p 1(\text{New}_{i,t}) \times 1(\text{Period}_p) + \sum_p \gamma_p 1(\text{Extant}_{i,t}) \times 1(\text{Period}_p) + \varepsilon_{i,t}.$$

Interpretation. These regressions show whether the composition of arriving depositors shifts toward insured, term, institutional, or listing-service accounts as the bank approaches failure.

3.4 Daily inflow regressions

The paper next studies daily inflow dynamics using regressions for the share and the level of new deposits. A representative specification for new-deposit levels is:

$$\log(\text{NewDeposits}_t) = \alpha + \sum_p \beta_p 1(\text{Period}_p) + \Gamma' Z_t + \rho \text{AR}(1)_t + \eta \text{RateSpread}_t + \varepsilon_t,$$

where Z_t includes macro-financial controls such as the VIX, GDP growth, housing starts, the daily S&P 500 return, and the OFR financial stress index.

Interpretation.

- The period dummies track whether inflows surge during bank-specific distress.
- The macro controls test whether inflows are driven only by aggregate conditions.
- The coefficient on **RateSpread** tests whether above-market pricing actively attracts new deposits.

3.5 Elasticity calculations for inflows

Using the estimated effect of the offered-rate spread on new deposits, the paper backs out several key elasticities:

- elasticity of the **new-deposit share**: about **1.52**;
- elasticity of the **quantity of new deposits**: about **3.05**;
- implied elasticity of **existing deposit balances**: about **0.61**.

Interpretation. Deposit pricing matters greatly even for a distressed bank. This is central for the paper's claim that insured inflows are rate-sensitive and therefore can be actively attracted.

3.6 Generalization to all U.S. banks under regulatory action

For bank j in quarter t , the paper studies funding-share outcomes using:

$$y_{j,t} = \alpha + \beta \text{UnderRegAction}_{j,t} + \gamma X_{j,t} + \delta_j + \zeta_t + \varepsilon_{j,t},$$

where $y_{j,t}$ is one of the following funding shares:

- brokered deposits,
- listing service deposits,
- small term deposits (\$100,000 or less),
- medium term deposits (\$100,001 to \$250,000),
- large term deposits (above \$250,000).

The dynamic event-study version is

$$y_{j,t} = \alpha + \sum_{i=-4}^{\geq 5} \beta_i \text{UnderRegAction}_{j,t=\tau+i} + \gamma X_{j,t} + \delta_j + \zeta_t + \varepsilon_{j,t}.$$

Interpretation. These regressions test whether the same shift from uninsured and brokered funding toward more insured retail-like term funding appears in the broader banking system when banks become distressed.

3.7 Large-bank rate-response regression

To test whether larger banks behave similarly, the paper estimates a rate-spread regression of the form

$$\text{CDSpread}_{j,t} = \alpha + \beta \text{UnderRegAction}_{j,t} + \zeta_t + \varepsilon_{j,t},$$

for 12-month, \$100,000 CDs.

Interpretation. This regression asks whether large banks under enforcement action increase rates enough to attract insured term deposits, just like the smaller failed bank in the microdata.

4 Identification and instruments (what makes the estimates credible)

4.1 What standard approaches miss

- Standard bank balance-sheet data mostly reveal **net deposit changes**, not separate gross inflows and gross outflows.
- A bank can therefore appear stable in aggregate even when uninsured depositors are running and new insured depositors are replacing them.
- The paper’s central empirical contribution is therefore a measurement contribution: it observes the funding mechanics that standard datasets suppress.

4.2 What identifies depositor stability in the microdata

- **Insurance variation.** The contrast between insured and uninsured balances identifies depositor sensitivity to bank distress.
- **Temporary guarantees.** The post-2008 introduction of TAG and DFA changes effective coverage and provides variation in coverage beyond the usual limit.
- **Account structure.** Checking, direct-deposit, trust, maturity, and prior activity help identify which kinds of accounts are sticky.
- **Timing of information.** The separation into Placebo, Precrisis, Postcrisis, and Formal periods distinguishes normal times, general financial stress, and severe bank-specific bad news.

4.3 What identifies inflows

- The paper observes the exact timing and composition of new accounts.
- The bank’s use of listing services lines up closely with inflow surges.
- Above-market offered rates help explain daily inflows even after controlling for macro-financial conditions.

- The clustering of balances just below the insurance limit supports the interpretation that the inflows are deliberately structured to exploit insurance.

4.4 What identifies the broader external evidence

- In the Call Report panel, the treatment variable is a bank-quarter indicator for regulatory action or closely related funding restrictions.
- The regressions control for bank health, growth, size, deposit reliance, and time and bank fixed effects.
- The event-study specification checks pretrends and the dynamic path around treatment.
- Separate RateWatch evidence on large-bank CD spreads shows that banks under enforcement action actively raise rates.

4.5 Why the design is credible

- The failed-bank evidence is extraordinarily granular and directly measures the funding flows of interest.
- The broader bank panel shows similar changes in funding composition, which increases external credibility.
- The paper is transparent about the main limitation: detailed deposit microdata were only usable for one bank, because comparable post-failure records for other banks were incomplete or could not be consistently linked.

4.6 Relation to the literature

- The paper speaks to the literature on **bank runs and depositor discipline**, but shows that discipline is weakened when banks can attract insured replacement funding.
- It also relates to the **deposit competition** literature by showing that rate-setting remains powerful even for distressed banks.
- More broadly, it contributes to the literature on **liquidity regulation and funding stability** by emphasizing that funding inflows must be studied alongside outflows.

5 Tables and figures

5.1 Figure 1 and Table I: bank background and summary statistics

- **Read Figure 1 as the buildup of stress.** Uninsured balances, especially uninsured term deposits, begin running already before the peak of the crisis and then accelerate sharply as the bank approaches failure.
- **Read Table I as a composition map.** By the Formal period:

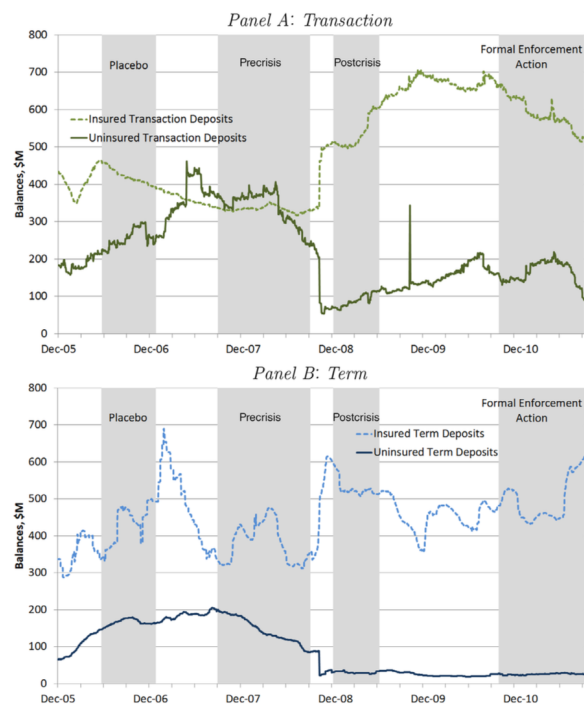


Figure 1. Deposit balances. This figure shows total balances in transaction (Panel A) and term (Panel B) deposit accounts. Gray bars denote the time periods analyzed in the regressions of Section II, and overlaid text identifies the name of each period. Note that the dramatic, brief spike in uninsured transaction deposits between the Postcrisis and Formal periods reflects a single transaction in which another subsidiary of the bank's holding company passed funds through the bank in such a manner that they remained within the bank for a few days.

Table I
Summary Statistics, by Period

Panel A reports summary statistics for new depositors arriving within the four event periods while Panel B reports statistics for depositors extant at the bank at the beginning of each period and at the date of failure. See Table II for precise variable definitions.

Panel A: New Depositors				
	Placebo (1)	Precrisis (2)	Postcrisis (3)	Formal (4)
Number of New Accounts	3,028	2,146	927	2,600
Uninsured at Start of Account	0.063	0.087	0.044	0.027
TAG/DFA Eligible	0.015	0.038	0.016	0.008
↔ (Covered in Postcrisis & Formal)				
Starting Balance	43,520	90,840	141,000	186,900
Term (count weighted)	0.461	0.478	0.417	0.824
Savings (count weighted)	0.487	0.379	0.408	0.084
Checking (count weighted)	0.052	0.144	0.175	0.092
Starting Interest Spread to Market	2.561	1.656	0.796	0.671
Institutional - Listed/Faxed	0	0.006	0.013	0.707
Institutional - Other	0.031	0.252	0.245	0.100
Brokered/Placed	0.002	0.035	0.221	0
Trust	0.038	0.037	0.094	0.041

Panel B: Extant Depositors					
	Placebo (1)	Precrisis (2)	Postcrisis (3)	Formal (4)	Failure (5)
Number of Accounts	41,366	44,910	38,374	30,342	25,141
Uninsured at Start of Period	0.065	0.087	0.009	0.021	0.014
TAG/DFA Eligible	0.006	0.008	0.004	0.004	0.005
↔ (Covered in Postcrisis & Formal)					
Starting Balance	28,310	27,710	31,450	43,970	49,750
Term (count weighted)	0.190	0.242	0.222	0.123	0.191
Savings (count weighted)	0.734	0.690	0.700	0.768	0.678
Checking (count weighted)	0.076	0.068	0.079	0.109	0.130
Starting Interest Rate	4.121	4.367	2.486	0.941	0.882
Starting Interest Spread to Market	2.604	2.576	1.492	0.673	0.632
Institutional - Listed/Faxed	0	0	0	0.002	0.084
Institutional - Other	0.013	0.015	0.025	0.047	0.065
Brokered/Placed	0.013	0.012	0.045	0.037	0.005
Trust	0.015	0.017	0.016	0.023	0.028
Direct Deposit	0.028	0.030	0.023	0.035	0.032
Past-Month Fees	-0.237	-0.061	0.084	0.174	0.171
Age of Relationship in Years	2.214	3.076	4.031	5.587	6.032
Years Since Start of Previous Period	-	1.25	1.25	1.78	0.92

- new depositors are far more concentrated in term deposits,
- average balances are much larger,
- institutional listed/faxed accounts dominate,
- uninsured exposure among new accounts is much lower.

- **Main takeaway.** The bank's liability structure did not simply shrink. It *changed*: uninsured and brokered-like funding ran, while insured institutional term funding arrived.

5.2 Table III, Table IV, Table VII, and Table VIII: who withdraws?

- **Transaction accounts.**
 - In the Formal period, uninsured transaction accounts liquidate about **18.3 percentage points** more often than insured accounts in the baseline LPM.
 - Checking accounts are substantially more stable.
 - Direct-deposit accounts are much more stable.
 - Older relationships are more stable.
- **Temporary insurance.** TAG/DFA eligibility reduces outflows during the crisis, especially in the postcrisis period.
- **Term accounts.**

Table III

Who Withdraws? Transaction Deposits: Linear Probability Model

This table presents estimates from linear probability models of the probability of account liquidation during each of the four periods of interest. Liquidation is defined as withdrawing 75% or more of the start-of-period account balance and staying at or below 25% of the start-of-period balance for at least 61 days. See Table II for additional variable definitions. *t*-Statistics are in parentheses, calculated using asymptotically normal standard errors. ***, **, and * indicate significance at 99%, 95%, and 90%, respectively. † TAG/DFA coverage was effective only in the Postcrisis and Formal periods; eligible accounts were fully covered/insured in those periods. In earlier periods, eligibility does not imply coverage and is shown for comparison purposes only.

	Placebo (1)	Precrisis (2)	Postcrisis (3)	Formal (4)
<i>Uninsured</i>	0.0249* (1.94)	-0.00117 (-0.09)	0.0693* (1.92)	0.183*** (8.45)
<i>TAG/DFA Eligible</i> †	-0.0446 (-1.50)	-0.000624 (-0.02)	-0.0944 ** (-2.00)	-0.0412 (-0.95)
<i>Checking</i>	-0.0901*** (-8.44)	-0.105*** (-8.10)	-0.0277*** (-2.64)	-0.0252** (-2.15)
<i>Direct Deposit</i>	-0.104*** (-7.08)	-0.123*** (-7.47)	-0.0864*** (-5.88)	-0.0459*** (-2.77)
<i>Log(Age)</i>	-0.0194*** (-7.47)	-0.0180*** (-4.34)	-0.00915** (-2.40)	-0.0247*** (-4.65)
<i>Prior Transactions</i>	0.00793*** (14.13)	0.00555*** (7.62)	0.00263*** (4.37)	-0.00310*** (-4.19)
<i>Prior Transactions</i> ²	-0.0000983*** (-12.71)	-0.0000746*** (-7.19)	-0.0000419*** (-4.88)	0.0000235** (2.34)
<i>Institutional - Any</i>	-0.0273 (-1.16)	0.0242 (1.01)	0.0218 (1.29)	0.0215 (1.33)
<i>Trust</i>	0.0255 (0.93)	0.0248 (0.83)	-0.0418* (-1.90)	0.0698*** (3.38)
Constant	0.217*** (13.12)	0.313*** (17.35)	0.159*** (11.62)	0.326*** (21.96)
Branch controls	Yes	Yes	Yes	Yes
No. of accounts	33,498	34,029	29,865	26,616
No. of liquidations	6,920	10,795	4,740	6,218
<i>R</i> ²	1.3%	0.8%	0.4%	1.8%
Model <i>p</i> -value	<0.001	<0.001	<0.001	<0.001

Table IV

Who Withdraws? Transaction Deposits: Alternative Specifications

This table presents estimates from Probit (columns (1) and (2)) and Cox proportional hazard models (columns (3) and (4)) of the probability of account liquidation during the Postcrisis and Formal periods. *Liquidation* is defined as withdrawing 75% or more of the start-of-period account balance and staying at or below 25% of the start-of-period balance for at least 61 days. See Table II for additional variable definitions. Probit estimates are expressed as marginal effects (with a coefficient of 0 reflecting a variable having no effect on the probability of liquidation), and Cox estimates are expressed as hazard ratios (with 1 indicating no effect). *t*-Statistics are in parentheses, calculated using asymptotically normal standard errors. ***, **, and * indicate significance at 99%, 95%, and 90%, respectively.

	Probit		Cox P.H.	
	Postcrisis (1)	Formal (2)	Postcrisis (3)	Formal (4)
<i>Uninsured</i>	0.0655 (1.61)	0.170*** (6.69)	1.503** (1.99)	1.862*** (8.01)
<i>TAG/DFA Eligible</i>	-0.0691** (-2.24)	-0.0164 (-0.42)	0.567* (-1.83)	0.958 (-0.26)
<i>Checking</i>	-0.0262*** (-2.71)	-0.0280** (-2.51)	0.814*** (-2.72)	0.887** (-2.15)
<i>Direct Deposit</i>	-0.0875*** (-8.55)	-0.0679*** (-4.37)	0.430*** (-6.04)	0.631*** (-4.32)
<i>Log(Age)</i>	-0.00798** (-2.18)	-0.0231*** (-4.49)	0.936*** (-2.63)	0.896*** (-4.72)
<i>Prior Transactions</i>	0.00290*** (4.61)	-0.00337*** (-4.33)	1.025*** (5.43)	0.985*** (-3.75)
<i>Prior Transactions</i> ²	-0.0000518*** (-5.15)	0.0000259** (2.36)	1.000*** (-5.56)	1.000* (1.68)
<i>Institutional - Any</i>	0.0226 (1.24)	0.0192 (1.18)	1.156 (1.27)	1.090 (1.24)
<i>Trust</i>	-0.0444** (-2.30)	0.0601*** (2.73)	0.711* (-1.92)	1.250*** (2.71)
Branch controls	Yes	Yes	Yes	Yes
No. of accounts	29,865	26,616	29,865	26,616
No. of liquidations	4,740	6,218	4,740	6,218
Log likelihood	-13,001.0	-14,236.2	-48,378.0	-62,351.7
Model <i>p</i> -value	<0.001	<0.001	<0.001	<0.001

- Uninsured term deposits are also more likely to liquidate in stressed periods.
- Accounts farther from maturity are much less likely to liquidate.
- Brokered/placed deposits are especially unstable.
- Newly arrived institutional listed/faxed term deposits are relatively sticky even late in the bank's life.
- **Uninsured migration.** Table VIII shows that uninsured transaction depositors often withdraw much *more* than needed to become fully insured; many either drop well below the limit or exit almost completely.
- **Main takeaway.** Insurance matters, but depositor heterogeneity matters too. Not all deposits are equally runnable.

5.3 Figure 2, Figure 3, and Table IX: new depositors and funding composition

- **Figure 2.** The bank's funding mix shifts away from brokered or fleeing wholesale balances toward institutional term deposits.
- **Figure 3.** New insured term deposits surge late in the bank's life. The total reaches nearly **\$400 million**, roughly a third of the deposit base and about equal to the amount of fleeing deposits.
- **Table IX.** Relative to extant depositors in normal times, new depositors in the Formal period are:
 - much less likely to be uninsured,
 - much more likely to be institutional listed/faxed,
 - much more likely to be CDs,

Table VII
Who Withdraws? Term Deposits: Alternative Specifications

This table presents estimates from Probit (columns (1) and (2)) and Cox proportional hazard models (columns (3) and (4)) of the probability of account liquidation during the Postcrisis and Formal periods. *Liquidation* is defined as withdrawing 75% or more of the start-of-period account balance and staying at or below 25% of the start-of-period balance for at least 61 days. See Table II for additional variable definitions. Probit estimates are expressed as marginal effects (with a coefficient of 0 reflecting a variable having no effect on the probability of liquidation), and Cox estimates are expressed as hazard ratios (with 1 indicating no effect). *t*-Statistics are in parentheses, calculated using asymptotically normal standard errors. ***, **, and * indicate significance at 99%, 95%, and 90%, respectively. ‡ Note that although TAG/DFA did not directly insure term deposits, it could do so indirectly. Accounts covered by TAG/DFA were excluded in calculating a depositor's total exposure to a bank when applying standard, limited-value deposit insurance, effectively freeing up insurance coverage to be applied to other, non-TAG/DFA-covered accounts.

	Probit		Cox P.H.	
	Postcrisis (1)	Formal (2)	Postcrisis (3)	Formal (4)
<i>Uninsured</i>	0.110* (1.89)	0.167*** (2.80)	1.637** (2.55)	1.746*** (2.99)
<i>TAG / DFA Eligible</i> ‡	0.00446 (0.03)	—	0.915 (−0.12)	—
<i>Log(Age)</i>	−0.00498 (−0.98)	−0.0156 (−1.52)	1.017 (0.81)	1.053* (1.92)
<i>Log(Days to Maturity)</i>	−0.214*** (−43.56)	−0.115*** (−15.28)	0.545*** (−56.75)	0.664*** (−28.43)
<i>Brokered / Placed</i>	0.337*** (17.44)	0.586*** (29.93)	3.678*** (21.11)	8.565*** (26.58)
<i>Institutional - Listed / Faxed</i>	−0.169 (−1.18)	−0.233*** (−3.35)	0.788 (−0.24)	0.465** (−2.14)
<i>Institutional - Other</i>	−0.0135 (−0.31)	−0.0498 (−1.15)	0.914 (−0.46)	1.239 (1.50)
<i>Trust</i>	−0.00735 (−0.24)	0.0147 (0.31)	0.985 (−0.11)	1.220 (1.31)
Branch controls	Yes	Yes	Yes	Yes
No. of accounts	8,509	3,726	8,509	3,726
No. of liquidations	2,559	1,736	2,559	1,736
Log likelihood	−3,767.9	−1,856.9	−21,333.4	−12,883.1
Model <i>p</i> -value	<0.001	<0.001	<0.001	<0.001

Table VIII
Uninsured Transaction Account Migration

For all transaction accounts that had a balance of \$2,000 less than the current deposit insurance limit or higher at the beginning of each period, this table shows their distribution into various account-size bins at the end of the period. Several bins are suppressed for disclosure reasons.

Panel A: Deposit Insurance Limit = \$100,000						
Bin range	<\$1	\$1–2,000	\$2,000–48,000	\$48,000–98,000	\$98,000–102,000	>\$102,000
Placebo	4.98%	9.08%	11.42%	10.32%	11.71%	52.59%
Postcrisis	7.71%	9.59%	9.88%	15.59%	16.35%	40.88%
Panel B: Deposit Insurance Limit = \$250,000						
Bin range	<\$1	\$1–2,000	\$2,000–123,000	\$123,000–248,000	\$248,000–252,000	>\$252,000
Postcrisis	—	6.37%	14.22%	13.73%	—	65.69%
Formal	20.79%	6.6%	22.3%	14.04%	8.05%	27.72%

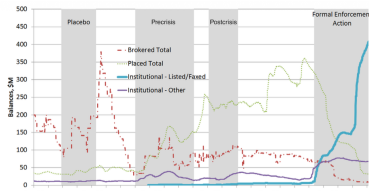


Figure 2. Term deposit balances in brokered, placed, and institutional accounts. This figure shows term deposit account balances in brokered accounts (dash-dotted red line), placed accounts (dotted green line), institutional deposits obtained via rate listing services and faxes (bold blue line), and other institutional deposits (bold, thin purple line). Placed deposits are those placed by a third party on behalf of the underlying depositor, where that third party does not meet the definition of deposit broker. Note that this is a different notion of placed deposits relative to that used in the regressions; here, we split placed and brokered deposits into two categories, whereas both were grouped as “placed” in the regressions. Among placed and brokered deposits, the underlying depositors are often not identified to the bank accepting the deposits. Gray bars denote the time periods analyzed in the regressions of Section II, and overlaid text identifies the name of each period.

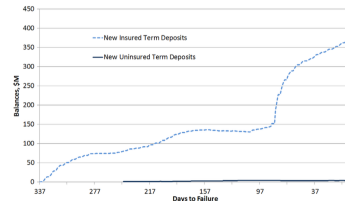


Figure 3. Term deposit balances from new depositors. This figure shows balances in term deposit accounts from depositors who opened their first deposit account with the bank after the formal enforcement action—new depositors.

Table IX
Comparison of Depositors across Periods

Each column corresponds to a separate account-level OLS regression of an account characteristic on the interaction of time period dummies and an indicator for extant or new depositors. The observational unit is account-time period, and with five time periods it is possible for an individual account to appear up to five times in this regression. Extant depositors are those with an existing deposit relationship at the beginning of each time period, and new depositors are those who arrive during the period. The omitted category is extant depositors in the Placebo period, so all estimates in other rows reflect differences relative to that category. *t*-Statistics are in parentheses, calculated using asymptotically normal standard errors. ***, **, and * indicate significance at 99%, 95%, and 90%, respectively. † TAG/DFA coverage was only effective in the Postcrisis and Formal periods; eligible accounts were fully covered/insured in those periods. In earlier periods, eligibility does not imply coverage and is shown for comparison purposes only.

Var. Type	Uninsured Dummy	TAG/DFA Eligible/ Dummy	Brokered/ Placed Dummy	Institutional: Listed / Faxed Dummy	Checking Dummy	CD Dummy
Panel A: Extant Depositors						
<i>Placebo</i>				Omitted Dummy		
<i>Postcrisis</i>	0.0212*** (15.29)	0.00205*** (3.82)	−0.000476 (−0.47)	−0.00000572 (−0.01)	−0.00803*** (−4.17)	0.0521*** (19.08)
<i>Postcrisis</i>	−0.0558*** (−38.71)	−0.00226*** (−4.04)	0.0025*** (30.92)	0.000240 (0.29)	0.00254 (1.27)	0.0315*** (11.11)
<i>Formal</i>	−0.0445*** (−28.91)	−0.00188*** (−3.15)	0.0242*** (21.64)	0.00197*** (2.23)	0.0228*** (15.37)	−0.0674*** (−22.27)
<i>Failure Date</i>	−0.0559*** (−31.27)	−0.00155*** (−2.46)	−0.00707*** (−5.97)	0.0843*** (90.07)	0.0543*** (24.02)	0.00108 (0.34)
Panel B: New Depositors						
<i>Placebo</i>	−0.00160 (−0.42)	0.00903*** (8.08)	−0.0105*** (−3.78)	−0.0000725 (−0.03)	−0.0242*** (−4.55)	0.271*** (35.92)
<i>Postcrisis</i>	0.0221*** (4.91)	0.0134*** (18.09)	0.0220*** (6.70)	0.00552*** (2.13)	0.0674*** (10.77)	0.297*** (32.42)
<i>Postcrisis</i>	−0.0208*** (−3.07)	0.0100*** (3.83)	0.209*** (42.42)	0.0129*** (3.11)	0.0987*** (10.31)	0.227*** (17.09)
<i>Formal</i>	−0.0377*** (−3.16)	0.00230 (1.44)	−0.0125*** (−4.18)	0.707*** (296.88)	0.0158*** (2.77)	0.634*** (75.30)
Constant	0.0650*** (64.97)	0.00616*** (15.90)	0.0125*** (17.20)	0.0000725 (0.13)	0.0781*** (54.73)	0.190*** (96.59)
No. of obs.	198,834	198,834	198,834	198,834	198,834	198,834
R ²	2.4%	0.3%	1.9%	35.0%	0.7%	5.2%
Model <i>p</i> -value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
F-statistic	572.9	61.32	463.5	12,798.7	161.3	1,295.1

- much less likely to be ordinary checking/savings accounts.

- **Main takeaway.** New money arriving at the failing bank was not random retail funding. It was mostly **insured institutional term money** sourced through listing services.

5.4 Figure 4, Table X, and Table XI: pricing and inflows

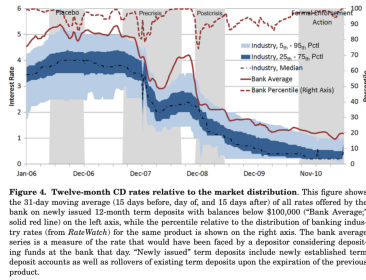


Table X
What Explains the Share of Deposits Which are New?

This table presents estimates from daily frequency regressions of the share of deposit dollars at the bank that are new (as of that day) on various controls. All models are OLS with Newey-West standard errors of lag length 9, chosen by the Newey-West rule of thumb (0.75√T) (Stock and Watson (2019)). ***, **, and * indicate significance at 99%, 95%, and 90%, respectively.

	(1)	(2)	(3)	(4)
Time period dummies:				
<i>Pre-Placebo</i>	5.439*** (2.90)	1.337 (1.13)	1.925* (1.65)	1.883* (1.65)
<i>Placebo to Precrisis</i>	3.027** (0.01)	2.054** (2.21)	3.079*** (3.05)	2.889*** (2.84)
<i>Precrisis</i>	-1.444*** (-2.72)	0.378 (0.25)	2.907 (1.58)	2.735 (1.49)
<i>Crisis</i>	0.188 (0.25)	2.009 (0.92)	3.417 (1.40)	3.157 (1.30)
<i>Postcrisis</i>	-0.393 (-0.40)	3.044 (0.38)	8.875** (2.20)	8.685** (2.15)
<i>Postcrisis to Formal</i>	-1.254** (-2.18)	2.230 (0.74)	9.364** (2.24)	8.919** (2.16)
<i>Formal</i>	5.821*** (2.79)	7.495** (2.14)	14.91*** (2.99)	14.46*** (2.90)
Economic controls:				
<i>LogVIX</i>	1.137 (0.63)	-1.363 (-0.51)	-0.806 (-0.32)	
<i>GDP Growth</i>	0.239** (2.21)	0.545*** (2.60)	0.514** (2.46)	
<i>Housing Starts</i>	0.00442 (1.57)	0.00415 (1.46)	0.00430 (1.47)	
<i>Daily S&P 500 Return</i>	18.19 (1.34)	13.93 (1.02)	15.04 (1.10)	
<i>AR(1)</i>	0.380*** (7.05)	0.379*** (7.04)	0.378*** (7.02)	
<i>OPR Financial Stress Index</i>	0.130 (0.60)	0.309* (1.74)	0.328 (1.48)	
<i>Rate Spread to Market (Dollar weighted average)</i>		5.549*** (2.71)	5.742*** (2.77)	
<i>Past-Month Fees/Deposit Balances (in Aggregate)</i>			1083.8 (0.36)	
Constant	3.218*** (6.62)	-5.345 (-1.62)	-12.91** (-2.55)	-13.45** (-2.62)
N	2,079	2,078	2,078	2,078
Model p-value	<0.001	<0.001	<0.001	<0.001

Table XI
What Explains the Log-Level of New Deposits?

This table presents estimates from daily frequency regressions of the logarithm of new deposit dollars at the bank (as of that day) on various controls. All models are OLS with Newey-West standard errors of lag length 9, chosen by the Newey-West rule of thumb (0.75√T) (Stock and Watson (2019)). ***, **, and * indicate significance at 99%, 95%, and 90%, respectively.

	(1)	(2)	(3)	(4)
Time period dummies:				
<i>Pre-Placebo</i>	1.152 (1.57)	0.410 (0.54)	0.673 (0.90)	0.662 (0.89)
<i>Placebo to Precrisis</i>	0.991 (1.46)	0.618 (0.88)	1.084 (1.51)	1.039 (1.39)
<i>Precrisis</i>	-1.822** (-2.57)	-0.123 (-0.11)	1.064 (0.95)	1.023 (0.90)
<i>Crisis</i>	-1.167 (-1.30)	1.307 (0.78)	1.963 (1.15)	1.902 (1.09)
<i>Postcrisis</i>	-1.068 (-1.17)	0.718 (0.38)	3.449* (1.67)	3.384 (1.63)
<i>Postcrisis to Formal</i>	-2.225*** (-3.40)	-0.540 (-0.46)	2.447 (1.16)	2.364 (1.09)
<i>Formal</i>	0.671 (0.95)	1.798 (0.98)	5.229** (2.44)	5.124** (2.32)
Economic controls:				
<i>LogVIX</i>	4.182*** (4.20)	3.055*** (2.72)	3.160*** (2.62)	
<i>GDP Growth</i>	-0.0225 (-0.25)	0.0706 (0.74)	0.0639 (0.64)	
<i>Housing Starts</i>	0.00178 (1.11)	0.00163 (1.01)	0.00167 (1.01)	
<i>Daily S&P 500 Return</i>	-8.577 (-0.65)	-10.48 (-0.78)	-10.14 (-0.75)	
<i>AR(1)</i>	0.276*** (18.61)	0.274*** (18.48)	0.274*** (18.42)	
<i>OPR Financial Stress Index</i>	-0.266*** (-2.81)	-0.155 (-1.29)	-0.167 (-1.28)	
<i>Rate Spread to Market (Dollar weighted average)</i>		2.696*** (3.15)	2.649*** (3.18)	
<i>Past-Month Fees/Deposit Balances (in Aggregate)</i>			251.5 (0.30)	
Constant	11.09*** (21.17)	-8.220** (-1.97)	-8.354** (-2.41)	-8.515** (-2.38)
N	2,079	2,078	2,078	2,078
Model p-value	<0.001	<0.001	<0.001	<0.001

- **Figure 4.** The bank's offered 12-month CD rate moved high in the market distribution. In the Formal period, the bank was often at a very high percentile relative to market peers.
- **Table X / Table XI.** New-deposit shares and new-deposit balances respond strongly to the offered spread relative to market.
- **Elasticity evidence.**
 - new-deposit share elasticity: about **1.52**,
 - new-deposit quantity elasticity: about **3.05**,
 - implied elasticity of existing balances: about **0.61**.
- **Main takeaway.** The late-life deposit surge was bank-driven, not just macro-driven. The bank offered unusually attractive rates and depositors responded.

5.5 Table XII, Figure 5, and Table XIII: generalization to other U.S. banks

- **Table XII.** Banks under regulatory action:
 - reduce **brokered deposits** by about **1.24 percentage points**,
 - increase **listing service deposits** by about **0.826 percentage points**,
 - increase **small** and **medium** term deposits,
 - reduce **large** term deposits above \$250,000.
- **Figure 5 and Table XIII.** The time path shows brokered deposits falling after treatment and listing-service / insured term deposits rising after treatment.

Table XII
Impact of Regulatory Action on Funding Shares For U.S. Banks

This table presents estimates from regressing funding shares on a regulatory action dummy and bank-level controls for all U.S. banks from 2000 to 2016. The regulatory action dummy is based on public and confidential supervisory data, other variables are from the regulatory filings of all U.S. banks. Dependent variables are expressed in percentage points. *NPLs*, refers to nonperforming loans and *A* is total assets. Observational units are bank-quarters, with brokered deposit data available from 2000 to 2016, listing service deposits from 2011 to 2016, and all other term deposit data from 2010 to 2016. *Small term deposits* are those with balances of \$100,000 or less, *medium term deposits* are between \$100,001 and \$250,000, and *large term deposits* are those over \$250,000. *t*-Statistics are in parentheses, calculated using asymptotically normal standard errors. ***, **, and * indicate significance at 99%, 95%, and 90%, respectively.

	Brokered (1)	Listed (2)	Sm. Term (3)	Med. Term (4)	Lg. Term (5)
<i>Under Reg. Action</i>	-1.24*** (-22.72)	0.828*** (11.40)	0.364*** (5.21)	0.466*** (6.25)	-0.830*** (-12.10)
<i>NPLs/A</i>	0.0102** (2.38)	0.00224 (0.42)	0.117*** (22.32)	-0.00903 (-1.62)	-0.108*** (-20.97)
% <i>ΔA</i>	0.0238*** (54.53)	0.0127*** (18.82)	-0.0092*** (-41.97)	0.0212*** (29.95)	0.0606*** (10.13)
<i>In(A)</i>	2.83*** (137.87)	0.354*** (8.31)	1.27*** (31.87)	-0.837*** (-19.66)	-0.432*** (-11.05)
<i>Deposits/A</i>	4.29*** (25.62)	1.62*** (7.44)	1.79*** (8.33)	1.81*** (7.82)	-3.60*** (-17.11)
<i>Term Deps/A</i>	14.4*** (173.14)	16.2*** (112.02)	29.1*** (211.50)	39.2*** (267.68)	31.6*** (234.54)
Constant	-41.9*** (-176.09)	-11.6*** (-22.44)	-3.58*** (-7.40)	4.15*** (8.94)	-8.575 (-1.21)
Bank & Quarter FE	Yes	Yes	Yes	Yes	Yes
<i>N</i>	554,180	162,123	193,306	193,306	193,306
<i>R</i> ²	13.0%	8.3%	68.6%	33.8%	24.1%
Model <i>p</i> -value	<0.001	<0.001	<0.001	<0.001	<0.001

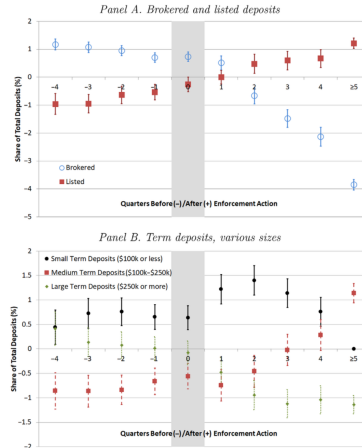


Figure 5. Impact of regulatory action on funding shares. This figure shows the impact of regulatory action on funding shares, using the estimates from Table XIII. *Small term deposits* are those with balances of \$100,000 or less, *medium term deposits* are between \$100,001 and \$250,000, and *large term deposits* are those over \$250,000. Time period *t* is the quarter in which the enforcement action was issued. Error bands represent 95% asymptotic confidence intervals.

Table XIII
Impact of Regulatory Action on Funding Shares For U.S. Banks, Quarterly Dummies

This table presents estimates from regressing funding shares on quarterly regulatory action dummies and bank-level controls for all U.S. banks from 2000 to 2016. The regulatory action dummies are based on public and confidential supervisory data, other variables are from the regulatory filings of U.S. banks. Dependent variables are expressed in percentage points. Observational units are bank-quarters, with brokered deposit data available from 2000 to 2016, listing service deposits from 2011 to 2016, and all other term deposit data from 2010 to 2016. *t*-Statistics are in parentheses, calculated using asymptotically normal standard errors. *Small term deposits* are those with balances of \$100,000 or less, *medium term deposits* are between \$100,001 and \$250,000, and *large term deposits* are those over \$250,000. Time period *t* is the quarter in which the regulatory action began. ***, **, and * indicate significance at 99%, 95%, and 90%, respectively.

	Brokered (1)	Listed (2)	Sm. Term (3)	Med. Term (4)	Lg. Term (5)
(<i>t</i> -4)	1.17*** (11.71)	-0.958*** (-5.00)	0.443*** (2.49)	-0.857*** (-4.51)	0.414*** (2.37)
(<i>t</i> -3)	1.07*** (11.12)	-0.961*** (-5.63)	0.728*** (4.68)	-0.863*** (-5.22)	0.138 (0.91)
(<i>t</i> -2)	0.951*** (10.30)	-0.633*** (-4.01)	0.759*** (5.20)	-0.838*** (-4.46)	0.0785 (0.56)
(<i>t</i> -1)	0.704*** (8.03)	-0.635*** (-3.78)	0.653*** (5.03)	-0.663*** (-4.77)	0.00887 (0.08)
(<i>t</i>)	0.729*** (8.33)	-0.260* (-1.91)	0.639*** (5.14)	-0.558*** (-4.21)	-0.0797 (-0.65)
(<i>t</i> +1)	0.512*** (4.03)	0.00124 (0.01)	1.22*** (7.90)	-0.743*** (-4.51)	-0.076*** (-3.14)
(<i>t</i> +2)	-0.667*** (-4.50)	0.477*** (2.72)	1.46*** (9.10)	-0.457*** (-2.77)	-0.945*** (-4.55)
(<i>t</i> +3)	-1.48*** (-12.53)	0.594*** (2.72)	1.14*** (5.66)	-0.0210 (-0.00)	-1.12*** (-7.08)
(<i>t</i> +4)	-2.13*** (-17.11)	0.698*** (4.43)	0.759*** (3.14)	0.284* (1.77)	-1.04*** (-7.08)
(<i>t</i> +1):2:5	-3.85*** (-41.34)	1.21*** (12.85)	-0.000122 (-0.00)	1.14*** (11.19)	-1.14*** (-12.18)
<i>NPLs/A</i>	0.0192*** (4.63)	0.00469 (0.94)	0.114*** (21.74)	-0.00804 (-1.43)	-0.106*** (-20.58)
% <i>ΔA</i>	0.0233*** (53.55)	0.0126*** (18.66)	-0.0098*** (-41.63)	0.0213*** (29.65)	0.0606*** (10.12)
<i>In(A)</i>	2.79*** (135.80)	0.383*** (8.96)	1.22*** (30.38)	-0.761*** (-17.78)	-0.455*** (-11.58)
<i>Deposits/A</i>	4.39*** (36.05)	1.55*** (7.13)	1.86*** (8.69)	1.80*** (7.21)	-3.32*** (-16.70)
<i>Term Deps/A</i>	14.4*** (173.11)	16.2*** (112.00)	29.0*** (211.38)	39.3*** (267.89)	31.6*** (234.61)
Constant	-41.4*** (-174.03)	-11.8*** (-22.86)	-3.01*** (-6.22)	3.40*** (6.56)	-8.384 (-1.81)
Bank & Quarter FE	Yes	Yes	Yes	Yes	Yes
<i>N</i>	554,180	162,123	193,306	193,306	193,306
<i>R</i> ²	13.9%	8.5%	68.8%	33.9%	24.1%
Model <i>p</i> -value	<0.001	<0.001	<0.001	<0.001	<0.001

Table XV
Large Bank Deposit Rate Responses to Enforcement Actions

This table presents estimates from pooled OLS regression of 12-month, \$100,000 CD rate spreads (measured against published FDIC national average rates) on a dummy indicating the bank is currently under regulatory enforcement action, time fixed effects, and a constant. Units are in percentage points: a value of 0.01 corresponds to 1 basis point (0.01%). The sample of banks is constrained to those over \$5 billion in assets (as of 2008Q4) that are under enforcement action or will be within four quarters. The coefficient on *Under Reg. Action* can therefore be interpreted as large banks' average change in deposit rates around/following the imposition of enforcement actions. *t*-Statistics are in parentheses, calculated using asymptotically normal standard errors. ***, **, and * indicate significance at 99%, 95%, and 90%, respectively.

	\$100K, 12m CD Spread
<i>Under Reg. Action</i>	0.4121*** (4.48)
Constant	0.3547*** (2.99)
Quarter FEs	Yes
<i>N</i>	65
<i>R</i> ²	6.9%
Model <i>p</i> -value	<0.001

- **Main takeaway.** The compositional substitution documented in the failed bank is visible in the broader U.S. banking system as well.

5.6 Table XV: large-bank rate responses

- Large banks under regulatory action raise 12-month \$100,000 CD spreads by about **41 basis points**.
- Together with the constant term, this implies that treated large banks offer rates almost exactly **75 basis points above** the national average, that is, close to the regulatory cap.
- **Main takeaway.** The high-rate / insured-inflow mechanism is not confined to one failed regional bank. Large banks under stress also use deposit pricing aggressively.

6 Short summary

- **Method.** Use daily account-level data from a failed bank to separately measure gross inflows and outflows; then generalize with all-bank panel regressions and deposit-rate evidence.

- **Identification.** Compare insured and uninsured accounts, exploit crisis-era insurance expansions, separate normal times from market-wide stress and bank-specific distress, and relate new-deposit inflows to above-market deposit rates.
- **Application.** One failed U.S. bank with unusually rich microdata, plus all U.S. banks facing regulatory action.
- **Main result.** Failing banks can lose uninsured deposits and still maintain funding through large inflows of newly originated insured term deposits, especially when they raise rates and target balances just below the insurance limit.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that **deposit inflows are first-order** in distressed banks.
- A failing bank can offset uninsured deposit outflows by attracting **new insured deposits**.
- Deposit insurance therefore stabilizes funding not only by preventing exits but also by making new entry of insured deposits possible.

7.2 Economic intuition

- Uninsured depositors discipline weak banks by running after bad news.
- But if the same bank can offer attractive rates to sophisticated depositors whose balances are fully insured, the bank can refill its balance sheet with protected funding.
- This makes **gross flows**, rather than net flows alone, central for understanding bank fragility.
- The mechanism is especially powerful when deposits can be structured just below the insurance limit and sourced nationally through listing services.

7.3 Takeaway

This paper's main lesson is simple but important: **a bank run is only half the story if one looks only at withdrawals**. In modern banking, especially under deposit insurance, distressed banks may simultaneously experience runs by uninsured depositors and inflows from new insured depositors. That fact changes how we should think about depositor rules, rate caps, brokered-deposit regulation, and the stability of bank funding more generally.